DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

236B Total Parameters

21B Activated / Token

128K Context Length

Multi-Head Latent Attention · DeepSeekMoE · 8.1T Token Pretraining

arXiv: 2405.04434

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

Training Cost

Scaling model size improves intelligence
but training FLOPs and GPU-hours grow
super-linearly, making large dense models prohibitively expensive.

KV Cache Bottleneck

Standard Multi-Head Attention stores full K/V vectors per head per token. At long contexts (128K), this dominates
GPU memory and limits batch size.

Inference Throughput

Dense models activate all parameters for every token. Serving a 70B dense model at scale requires massive hardware, driving up deployment cost.

DeepSeek-V2's answer: Combine Multi-Head Latent Attention (MLA) to compress KV cache with DeepSeekMoE to sparsely activate only 21B of 236B parameters — achieving top-tier quality at a fraction of the cost.

DeepSeek-V2 Architecture

MLA + DeepSeekMoE in a Transformer Block

Multi-Head Latent Attention (MLA)

- Replaces standard MHA in attention layers
- Low-rank joint compression of K and V
- Caches compact latent vector c_t^KV
- Reconstructs full K, V on-the-fly via projections
- Decoupled RoPE for positional encoding
- 93.3% KV cache reduction vs. MHA

DeepSeekMoE (FFN Replacement)

- Replaces dense FFN in each block
- Shared experts (N_s): always active
- Routed experts (N_r): sparse top-K activation
- Fine-grained segmentation → specialization
- Device-limited routing for comm. efficiency
- 236B total params, only 21B activated/token

Each Transformer block: Input → MLA → DeepSeekMoE FFN → Output (see paper Figure 1)

Multi-Head Latent Attention (MLA)

Compresses KV Cache by 93.3% While Outperforming MHA

MHA

Baseline

Full K,V per head
per token cached

High quality
High cache cost

GQA

Compromise

K,V shared across
groups of heads

Reduced cache
Some quality loss

MQA

Aggressive

Single K,V shared
across all heads

Minimal cache
Larger quality loss

MLA

DeepSeek-V2

Low-rank latent
vector c_t^KV cached

93.3% less cache
Better than MHA

KEY INSIGHT

MLA compresses K and V into a shared low-rank latent vector and reconstructs full attention on-the-fly. Decoupled RoPE handles positional encoding separately. Result: strictly better than MHA in quality, with 93.3% less KV cache — making GQA and MQA unnecessary.

DeepSeekMoE Architecture

Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

- Always activated for every token
- Capture common, shared knowledge
- Provide stable base representations

Routed Experts (N_r)

- Sparsely activated via top-K router
- Fine-grained segmentation for specialization
- Only a subset activated per token

Device-Limited Routing

Each token routed to experts on at most M devices → minimal cross-node communication

Auxiliary Load Balancing

Loss terms enforce balanced expert utilization across devices and batches

Token-Dropping Strategy

Prevents expert overflow during training; no tokens dropped at inference time

Efficiency Gains: DeepSeek-V2 vs. DeepSeek 67B

42.5% Training Cost Reduction and 5.76× Generation Throughput

DeepSeek 67B (Dense Baseline)

Architecture	**Dense Transformer**
Total Parameters	**67B**
Activated Parameters	**67B (all)**
Training Cost	**Baseline (1.0×)**
KV Cache	**Baseline (1.0×)**
Generation Throughput	**Baseline (1.0×)**
MMLU	**73.0**

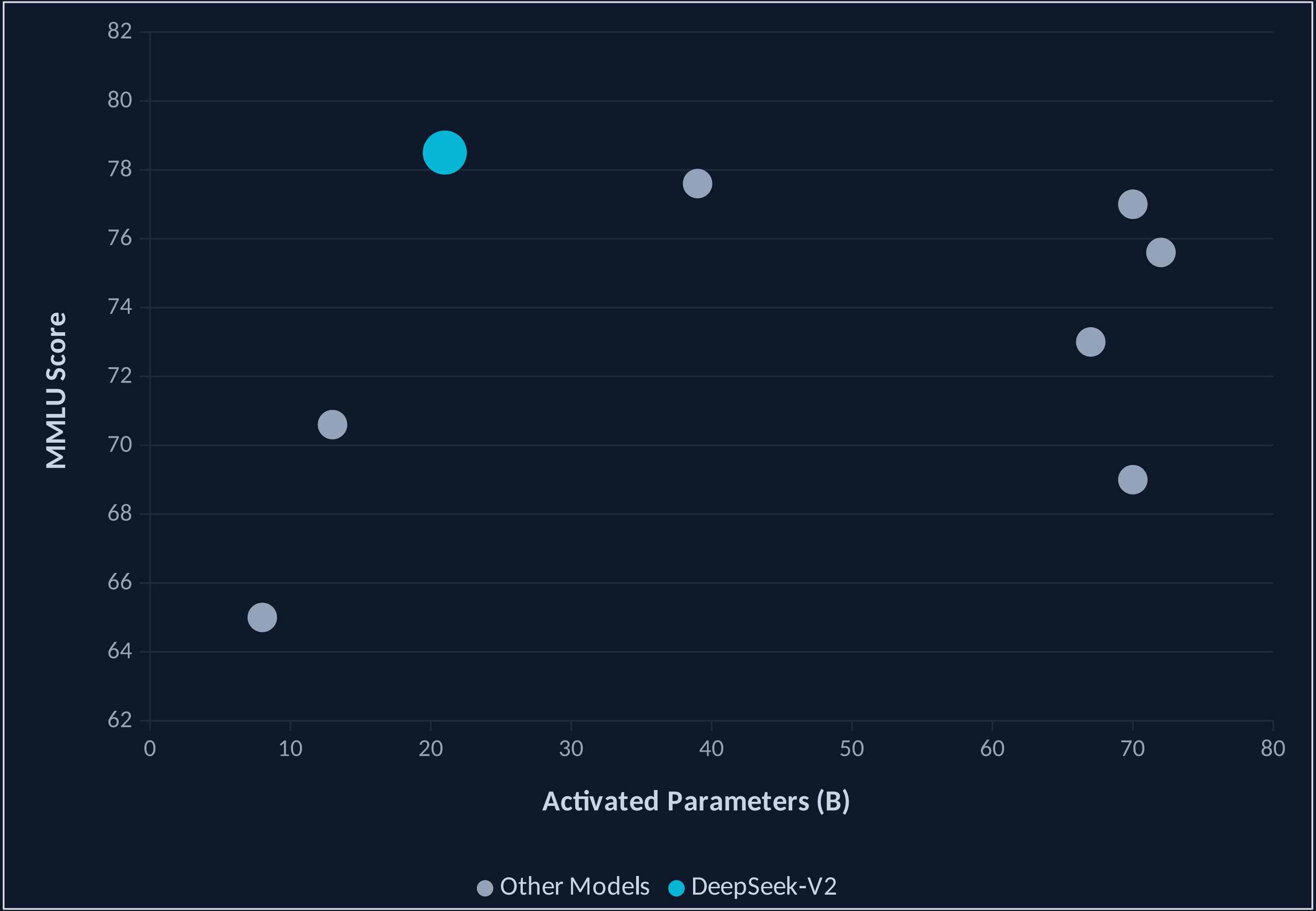
DeepSeek-V2 (MLA + MoE)

Architecture	**MLA + DeepSeekMoE**
Total Parameters	**236B**
Activated Parameters	**21B (8.9% of total)**
Training Cost	**42.5% lower**
KV Cache	**93.3% smaller**
Generation Throughput	**5.76× higher**
MMLU	**78.5 (+5.5 pts)**

DeepSeek-V2 is cheaper to train, uses far less memory at inference, and serves tokens 5.76× faster — all while scoring higher on MMLU.

MMLU vs. Activated Parameters

DeepSeek-V2 Achieves Pareto-Optimal Efficiency Among Open-Source Models



MODEL DATA

LLaMA 3 70B
70B activated | MMLU 77.0

Mixtral 8×22B
39B activated | MMLU 77.6

Qwen1.5 72B
72B activated | MMLU 75.6

DeepSeek 67B
67B activated | MMLU 73.0

LLaMA 2 70B
70B activated | MMLU 69.0

Mixtral 8×7B
13B activated | MMLU 70.6

LLaMA 3 8B
8B activated | MMLU 65.0

DeepSeek-V2
21B activated | MMLU 78.5 ★ Pareto-optimal

Alignment & Chat Evaluation

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models

ALIGNMENT PIPELINE

1. Pretraining — 8.1T tokens multi-source corpus → **2. SFT** — 1.5M conversational sessions → **3. GRPO RL** — Group Relative Policy Optimization

AlpacaEval 2.0

38.9%

Length-Controlled Win Rate

MT-Bench

8.97

Overall Score (out of 10)

AlignBench

7.91

Overall Score (Chinese)

On AlignBench (Chinese), DeepSeek-V2 Chat (RL) outperforms all open-source models and most closed-source models — demonstrating exceptional multilingual alignment quality.

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

1. **MLA: Superior KV-Cache Solution**

   Multi-Head Latent Attention achieves better performance than MHA while reducing KV cache by 93.3% — making GQA and MQA obsolete for inference efficiency.

2. **DeepSeekMoE: Economical Training**

   Fine-grained experts with shared expert isolation enable training a 236B-parameter model at 42.5% lower cost than a 67B dense model.

3. **Top-Tier Quality at a Fraction of Activated Params**

   Only 21B activated parameters match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU and other benchmarks.

4. **5.76× Throughput for Practical Deployment**

   Massive inference speedup over DeepSeek 67B makes DeepSeek-V2 viable for large-scale, cost-effective serving.

5. **Strongest Open-Source Chinese Chat Model**

   After SFT + GRPO RL, DeepSeek-V2 Chat (RL) outperforms all open-source and most closed-source models on Chinese alignment benchmarks.

DeepSeek-V2 is Pareto-optimal: highest quality per activated parameter, lowest cost per quality point.